

Waveform-Wizard: A free tool for Spectro-Temporal visualization of Speech

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I. ABSTRACT

This paper introduces a MATLAB-like speech-transformation visualization tool designed to facilitate the analysis and exploration of speech data through comprehensive visual representations. The tool offers a user-friendly graphical interface (GUI) that enables users to examine various speech properties and transformations in detail. Key functionalities include plotting audio signals, analyzing zero-time windowing spectrograms (ZTWS), spectral flatness features (SFF), S-transform, Constant-Q analysis, formant peak identification, pitch contour visualization, Gammatone analysis, and voice activity detection. Users can compare multiple audio files simultaneously, with the flexibility to adjust plot sections according to their preferences.

II. INTRODUCTION

Speech transformation and visualization tools play a pivotal role in analyzing complex speech data, enabling researchers and practitioners to understand various acoustic features and transformations better. While MATLAB [1] provides robust capabilities for visualizing speech properties such as zero-time windowing (ZTW) [2], Single frequency filtering (SFF) [3], spectral flatness features [4], S-transform [5], and formant peak identification [6], it also comes with notable shortcomings.

Licensing constraints posed a significant hurdle. As MATLAB is commercial software that requires a license for full functionality, many potential users found it difficult to access and modify the application to suit their specific needs. Although the free MATLAB Runtime allowed users to run the application, it did not provide the flexibility to explore or extend its capabilities without a licensed version of MATLAB.

The graphical user interface (GUI) also needed improvement. The design and aesthetics of the MATLAB-generated interface did not meet the expectations of many users, who found it unintuitive and less convenient. Furthermore, optimizing the application's performance was challenging, as MATLAB did not offer the same efficiency as lower-level programming languages for certain tasks, necessitating additional effort to enhance code performance.

In response to these challenges, we developed a new cross-platform application using Python. The objectives of this redevelopment were multifold: to ensure platform independence, enhance the user interface, and provide comprehensive

documentation and support for users. The new application introduces several additional features to improve functionality and user experience. These include an intuitive GUI built with PyQt5, support for multiple graph types and simultaneous file analysis, the ability to save and resume analyses, and the ability to export visualizations in various formats. Users can now open '.wav' files directly from their native file browser and analyze specific audio segments through a simple drag-and-select feature.

To implement these enhancements, we meticulously mapped MATLAB functions to their Python equivalents using libraries such as SciPy, NumPy, and LibROSA, translating any unmatched functions into Python. The GUI was developed using PyQt5, integrating the newly implemented features with a focus on usability and performance. Extensive testing and optimization ensured that the application met the desired standards of functionality and stability. The final product was packaged using PyInstaller, facilitating seamless installation on Windows and Ubuntu systems, with continuous integration and deployment managed through GitHub Actions.

This paper details the methodology employed in transitioning from MATLAB to Python, the enhancements to the application's functionality, and the performance improvements achieved, providing a comprehensive overview of the new speech transformation and visualization tool.

III. APPLICATION OVERVIEW AND FUNCTIONALITY

The **Waveform-Wizard** is a comprehensive speech-transformation visualization tool designed to empower users with detailed visual analysis capabilities. This section provides an overview of the application's functionality, supported by descriptions to guide users through its features.

A. Core Functionalities

- **Audio Signal Plotting:** Users can load audio files in various formats and visualize their waveform representations. This feature gives a quick overview of the signal's amplitude over time.
- **Zero-Time Windowing Spectrogram (ZTWS):** This spectrogram provides a time-frequency representation using a zero-time windowing approach, offering insights into frequency content variations over time.
- **Spectral Flatness Feature (SFF):** SFF is visualized to analyze the tonal versus noisy characteristics of the

speech signal, providing an essential measure for audio signal analysis.

- **S-Transform:** A powerful tool for time-frequency analysis, the S-transform combines the benefits of wavelet transforms and short-time Fourier transforms, allowing users to explore the temporal and spectral characteristics simultaneously.
- **Constant-Q Analysis** [7]: This analysis provides a logarithmic frequency representation, particularly useful for musical and speech signals where pitch perception is logarithmic.
- **Formant Peaks Identification:** Essential for speech analysis, this feature identifies and visualizes formant frequencies, aiding in analysing vowel sounds and speech articulation.
- **Pitch Contour Visualization:** Pitch tracking is critical for prosody analysis. Our application provides a clear visual representation of pitch variations over time.
- **Gammatone Analysis** [8]: This feature uses a filter bank model based on the human auditory system, providing insights into how different frequencies are processed by the human ear.

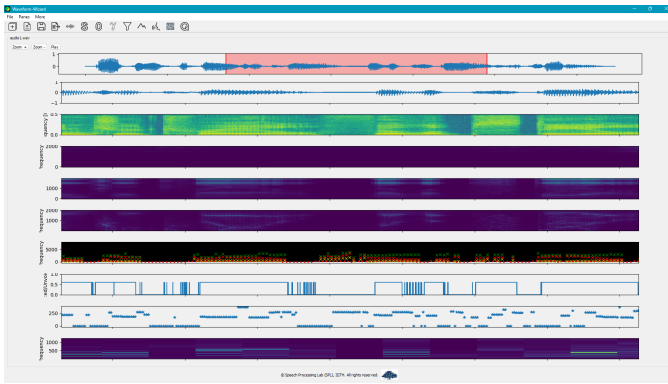


Fig. 1. All core functionality waveform representations

- **Voice Activity Detection (VAD)** [9]: VAD segments the audio signal into speech and non-speech regions, facilitating further processing and analysis of speech-active parts.

B. User Interface

The application features a user-friendly GUI built with PyQt5, designed to make navigation intuitive. Key elements include:

- **File Management:** Users can load, save, and export analyses in various formats, including PDF, PNG, and SVG. The GUI allows direct file opening from the native file browser, streamlining the workflow.
- **Pane Management:** Users can add or remove panes to view different analysis types simultaneously. Each pane corresponds to a specific visualization, such as a waveform, spectrogram, or pitch contour.
- **Interactive Analysis:** A drag-and-select feature enables users to focus on specific audio segments, enhancing

detailed analysis. All panes update dynamically to reflect the selected segment.

- **Customization:** Users can set custom titles and labels for their analyses, enhancing the clarity and presentation of exported visualizations.

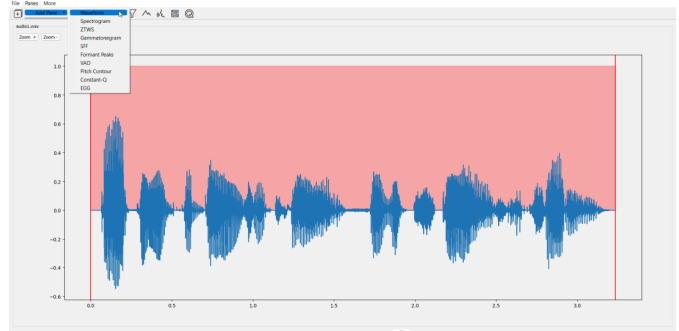


Fig. 2. All visualization options, user can select one or multiple visualizations and can select and view a portion of the signal

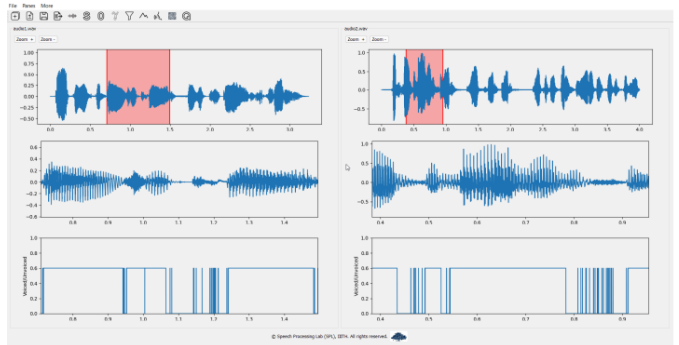


Fig. 3. Application Visualization

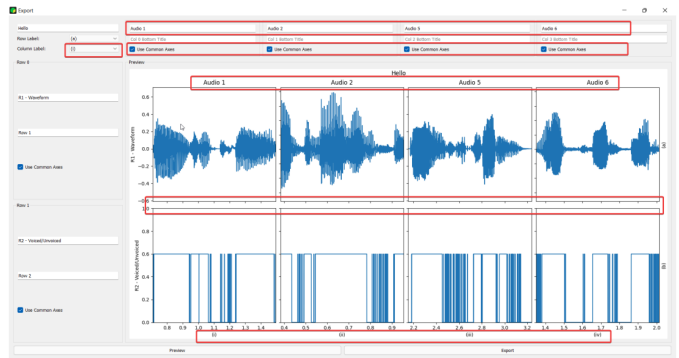


Fig. 4. A Picture of the export functionality

C. Application Workflow

- 1) **Loading Files:** Users start by selecting an audio file via the File -> Load File menu, opening a file picker to navigate and select the desired audio file.
- 2) **Adding Analysis Panes:** Through the Panes menu, users can add different analysis panes, each offering a unique perspective on the audio data.

- 3) **Comparing Multiple Files:** The application supports loading multiple files in the same session, allowing users to compare and contrast different audio signals side by side.
- 4) **Saving and Resuming Analysis:** Users can save their current analysis state and resume it later. The application uses a custom `.wwc` file format for storing analysis states.
- 5) **Exporting Results:** After completing the analysis, users can export their visualizations using the `File -> Export` menu, selecting their preferred format and destination.

This comprehensive walkthrough is designed to provide users with a detailed understanding of the capabilities of the **Waveform-Wizard**, enabling effective and insightful analysis of speech data. For a more in-depth explanation and demonstration, refer to the video.¹ To explore and try the platform, visit².

IV. CONCLUSION AND FUTURE WORK

In this paper, we introduced **Waveform-Wizard**, a Python-based speech-transformation visualization tool designed to enhance the analysis of speech data through detailed and intuitive visual representations. The transition from a MATLAB-based application to a Python-based solution addressed several critical issues, including performance optimization, licensing constraints, and user interface design.

The new application leverages Python's robust libraries and the PyQt5 framework to deliver a platform-independent, efficient, and user-friendly experience. With features such as zero-time windowing spectrograms, spectral flatness features, S-transform, and formant peak identification, **Waveform-Wizard** provides a comprehensive suite of tools for speech analysis. Additional functionalities, such as multi-file comparison, drag-and-select analysis, and customizable export options, further enhance its utility and flexibility.

Our extensive testing and optimization processes ensured the application's stability and performance across different operating systems. By incorporating user feedback, we significantly improved the GUI, making it more intuitive and aesthetically pleasing. The capability to create standalone applications using the MATLAB Runtime for deployment ensures broader accessibility while maintaining the application's core functionality.

Waveform-Wizard represents a significant advancement in speech data visualization tools, offering researchers and practitioners a powerful, flexible, and user-centric platform for exploring and analyzing complex speech properties. Future work will focus on further expanding the tool's capabilities, integrating additional analysis methods, and continuously refining the user experience based on evolving user needs and technological advancements.

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¹<https://www.youtube.com/watch?v=-EHPcCISOpU>

²<https://spl-btp-2k24.github.io/Waveform-Wizard/>